

Benjamin  
Karachun

Biotechnology & Food Engineering  
Software Development  
Applied AI

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00 PROFILE

Biotechnology and Food Engineering student at the Technion, specializing in computational biotechnology. Builds browser-based tools for scientific computing, education, and data workflows using Python, TypeScript/JavaScript, React, SQLite, and WebAssembly.

01 EXPERIENCE

Python Tutor Technion Student Community

- Tutored undergraduates in the Technion's Python programming course.
- Built and maintained TechnionPrep's Python study hub for course materials and exercises.

Technology Consultant Hanan Shemesh Accounting

- Integrated the WhatsApp Business API to automate client requests and routine communication.
- Digitized legacy records and introduced a searchable operational workflow.

Web Developer Aviator.co.il

- Maintained production WordPress infrastructure, managed updates and assets, and resolved operational issues.

02 SELECTED PROJECTS

MathPaster Chrome extension + web editor

mathpaster.com ↗

Built a MathLive-based WYSIWYG editor with autocomplete, matrix tools, history, and direct LaTeX insertion into AI chat interfaces. Equation data stays local to the browser.

TechnionPrep Next.js · React · TypeScript

bfe.technionprep.com ↗

Built and deployed a semester-by-semester study hub that organizes course materials, summaries, and interactive tools for Technion Biotechnology and Food Engineering students.

Shoresh Python ETL · SQLite · WebAssembly

Open project ↗

Developed a Python ETL pipeline for Israeli Ministry of Health food data and a browser-based SQLite application for nutrition search, comparison, recipes, and label generation.

Python Code Tester Pyodide · Web Workers · JSZip

Open project ↗

Built a private, in-browser homework tester that runs uploaded Python solutions against input/output cases locally, with bulk folder and ZIP import and structured result comparison.

03 EDUCATION

Technion — Israel  
Institute of Technology

B.Sc. Biotechnology & Food Engineering

Expected 2027 · Haifa, Israel

Computational Biotechnology, Food Engineering & Health

04 TECHNICAL

LANGUAGES

Python, TypeScript/JavaScript, SQL

FRONTEND

React, Next.js, HTML/CSS

DATA & RUNTIMES

NumPy, Pandas, SQLite,  
WebAssembly, Pyodide

PLATFORMS

Firebase, REST APIs, Chrome Extensions

TOOLING

Git, Docker, Linux/Bash

05 APPLIED AI

- OpenAI, Gemini, and Claude API integration
- Structured JSON outputs and schema validation
- Prompt and evaluation workflows
- RTL/LTR and MathJax content pipelines

06 LABORATORY

PCR, gel electrophoresis, aseptic technique, plate culture and counting, 96-well plate readers, and spectrophotometry.

07 LANGUAGES

|         |        |
|---------|--------|
| Hebrew  | Native |
| English | Fluent |
| Russian | Fluent |